

Microsoft Azure

AI-900

AI Fundamentals

Complete Study Guide

Covers all exam domains with visual explanations

Domain	Topic	Exam Weight
1	AI Concepts & Workloads	15–20%
2	Machine Learning Principles	20–25%
3	Computer Vision	15–20%
4	Natural Language Processing	15–20%
5	Generative AI	15–20%
6	Responsible & Trustworthy AI	10–15%

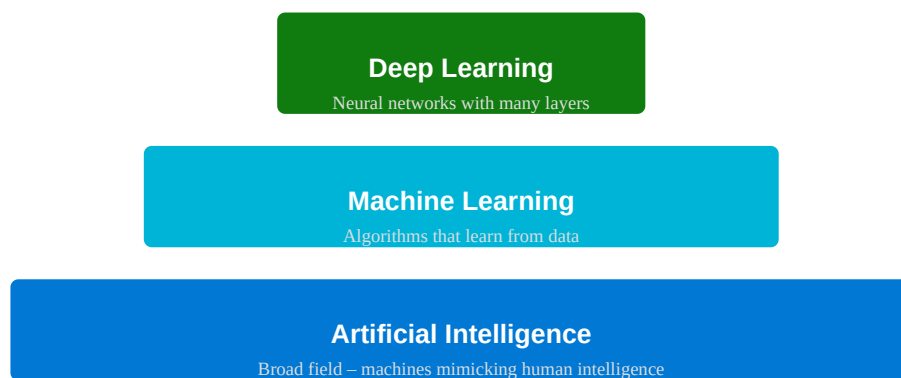
Domain 1: AI Concepts & Workloads

What is AI and what can it do?

What is Artificial Intelligence?

Artificial Intelligence (AI) is the simulation of human intelligence processes by computer systems. AI enables machines to perform tasks that typically require human intelligence — such as visual perception, speech recognition, decision-making, and language translation.

The AI Hierarchy



AI contains Machine Learning, which in turn contains Deep Learning.

Common AI Workloads

- **Machine Learning** – Training models to make predictions from data
- **Computer Vision** – Analysing and interpreting images and video
- **Natural Language Processing (NLP)** – Understanding and generating human language
- **Document Intelligence** – Extracting structured data from documents
- **Knowledge Mining** – Finding insights in large volumes of unstructured data
- **Generative AI** – Creating new content (text, images, code)

Term	Description
Prediction	A model output – e.g. 'this email is spam'
Training	The process of fitting a model to historical data
Inference	Using a trained model to make predictions on new data
Feature	An input variable used by the model (e.g. age, income)
Label / Target	The output variable the model is trying to predict

Domain 2: Machine Learning Principles

How machines learn from data

Types of Machine Learning

Supervised

Labelled training data
Regression & Classification

Unsupervised

No labels
Clustering & Anomaly detection

Reinforcement

Agent + Reward signal
Learn via trial & error

Regression vs Classification

Regression

Predicts a continuous number
e.g. "House price = £320,000"



Classification

Predicts a category/class
e.g. "Spam" or "Not Spam"



Key ML Concepts

Term	Description
Overfitting	Model memorises training data but fails on new data
Underfitting	Model is too simple to capture patterns in the data
Cross-validation	Technique to evaluate model by splitting data multiple ways
Confusion Matrix	Table showing True/False Positives and Negatives
Precision	Of all predicted positives, how many were correct?
Recall	Of all actual positives, how many were found?
F1 Score	Harmonic mean of precision and recall
ROC / AUC	Curve measuring classifier performance across thresholds
RMSE	Root Mean Square Error – common regression metric

Azure Machine Learning Studio

- **Automated ML (AutoML)** – Automatically tries many models and picks the best
- **Designer** – Drag-and-drop pipeline builder
- **Notebooks** – Full Python/R environment in the browser

- **Compute Clusters** – Scalable training compute
- **Endpoints** – Deploy models as REST APIs
- **Pipelines** – Reusable ML workflows

Domain 3: Computer Vision

Teaching machines to see

Computer Vision Pipeline



Data flows from raw image → pre-processing → feature extraction → inference → labelled output.

Computer Vision Tasks

Term	Description
Image Classification	Assign a label to the whole image (e.g. 'cat')
Object Detection	Locate and label multiple objects with bounding boxes
Semantic Segmentation	Classify every pixel in the image
Instance Segmentation	Distinguish individual object instances pixel-by-pixel
Optical Character Recognition (OCR)	Read printed or handwritten text from images
Facial Detection	Locate faces in an image
Facial Recognition	Identify or verify a person from their face

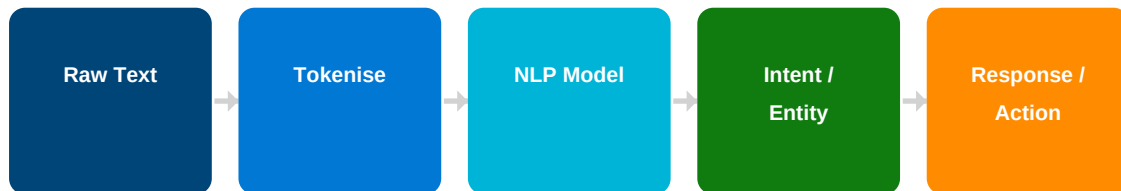
Azure Computer Vision Services

- **Azure AI Vision** – Image analysis, OCR, spatial analysis
- **Azure AI Face** – Face detection, verification, identification
- **Azure AI Custom Vision** – Train your own image classifier or detector
- **Azure AI Video Indexer** – Extract insights from video files
- **Azure AI Document Intelligence** – Extract structured data from forms & invoices

Domain 4: Natural Language Processing

Understanding human language

NLP Pipeline



Text is tokenised, processed by an NLP model, then intents/entities are extracted to drive a response.

NLP Tasks

Term	Description
Tokenisation	Splitting text into words or sub-words
Part-of-Speech (POS)	Labelling each word as noun, verb, adjective, etc.
Named Entity Recog.	Identifying people, places, dates, organisations in text
Sentiment Analysis	Detecting positive, negative, or neutral tone
Language Detection	Identifying which language text is written in
Translation	Converting text from one language to another
Summarisation	Generating a concise summary of longer text
Question Answering	Finding answers to questions within a text passage
Intent Recognition	Understanding what a user wants to accomplish
Entity Extraction	Pulling key information (dates, amounts, names) from text

Azure NLP Services

- **Azure AI Language** – Sentiment analysis, NER, summarisation, QA
- **Azure AI Translator** – Real-time text and document translation (100+ languages)
- **Azure AI Speech** – Speech-to-text, text-to-speech, speaker recognition
- **Azure AI Bot Service** – Build and connect intelligent chatbots
- **CLU (Conversational Language Understanding)** – Intent & entity for bots
- **Azure OpenAI** – GPT models for advanced language tasks

Domain 5: Generative AI

AI that creates new content

How Generative AI Works



A user prompt is tokenised, processed by a transformer model, decoded and returned as generated output.

Key Concepts

Term	Description
Large Language Model (LLM)	A neural network trained on vast text data to generate language
Transformer Architecture	Architecture using self-attention; powers GPT, BERT and others
Token	A chunk of text (word or sub-word) the model processes
Prompt	The input text given to a generative AI model
Prompt Engineering	Crafting prompts to get better, more accurate outputs
Temperature	Controls randomness: low = predictable, high = creative
Context Window	Maximum number of tokens a model can process at once
Grounding	Connecting LLM output to real, verifiable data sources
RAG (Retrieval-Augmented Generation)	Enhances LLM responses using retrieved documents
Fine-tuning	Further training a pre-trained model on domain-specific data
Embedding	A numeric vector representation of text used for search/similarity

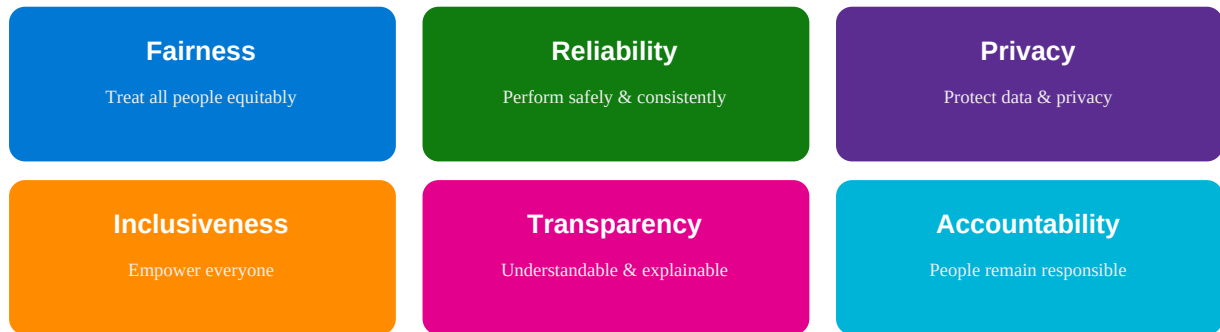
Azure Generative AI Services

- **Azure OpenAI Service** – Access to GPT-4, GPT-4o, DALL-E, Whisper via Azure
- **Azure AI Studio** – Central hub to build, evaluate and deploy AI apps
- **Microsoft Copilot** – AI assistant integrated across Microsoft 365
- **AI Search + RAG** – Ground LLM answers in your own data

Domain 6: Responsible & Trustworthy AI

Building AI ethically

Microsoft's 6 Principles of Responsible AI



Term	Description
Fairness	AI must treat all individuals equitably, avoiding bias based on age, gender, race, etc.
Reliability & Safety	AI must perform consistently and safely, even in unexpected situations
Privacy & Security	AI must protect personal data and resist adversarial attacks
Inclusiveness	AI should be accessible and beneficial to people of all abilities and backgrounds
Transparency	AI decisions should be explainable and understandable to users
Accountability	Humans retain responsibility for AI systems and their impacts

Responsible AI Tools on Azure

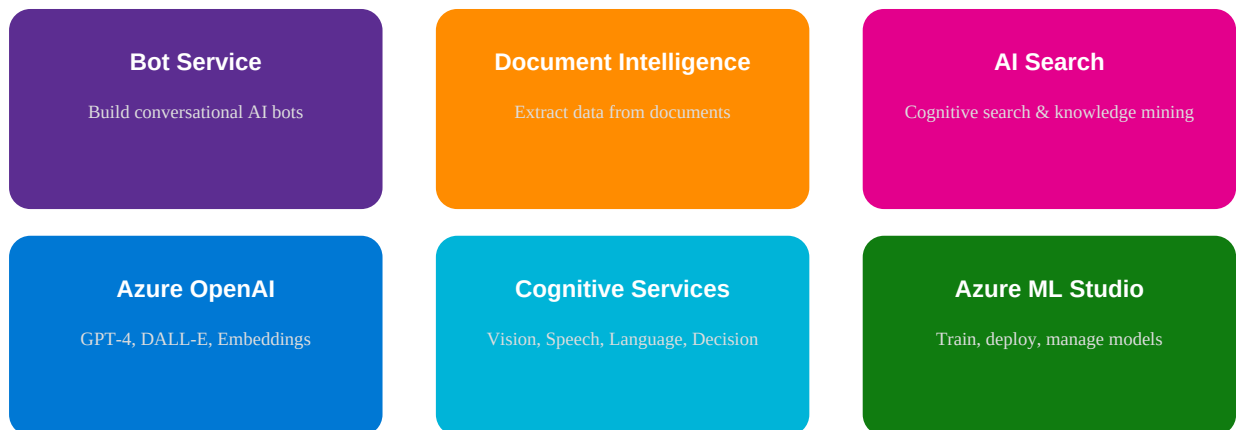
- **Azure AI Content Safety** – Detect harmful content in text and images
- **Fairlearn** – Open-source toolkit to assess and improve model fairness
- **InterpretML** – Explain model predictions (glass-box & black-box models)
- **Azure ML Responsible AI Dashboard** – Unified view of error analysis, fairness, explainability
- **Microsoft Responsible AI Standard** – Internal governance framework

Exam Tips

Tip	Detail
Tip	Detail
Know the 6 principles	Be able to match a scenario to the correct principle
Azure services mapping	Know which Azure service handles each AI task
ML metrics	Understand precision, recall, RMSE and when to use each
Generative AI basics	Know what LLMs, prompts, tokens and RAG mean
Responsible AI tools	Content Safety, Fairlearn, InterpretML are frequently tested

Azure AI Services at a Glance

Quick reference



Term	Description
Azure AI Vision	Image analysis, OCR, object detection, spatial analysis
Azure AI Face	Face detection, verification, identification
Azure AI Custom Vision	Train custom image classifiers and object detectors
Azure AI Language	Sentiment, NER, summarisation, Q&A, CLU
Azure AI Translator	Text/document translation in 100+ languages
Azure AI Speech	STT, TTS, speaker recognition, live captions
Azure AI Document Intelligence	Form & invoice data extraction
Azure AI Bot Service	Build, test, deploy conversational bots
Azure OpenAI Service	GPT-4, GPT-4o, DALL-E 3, Whisper, Embeddings
Azure AI Studio	Unified portal to build and manage AI projects
Azure Machine Learning	Train, track, deploy and monitor ML models
Azure AI Search	Full-text + vector search; supports RAG patterns
Azure AI Content Safety	Detect hate, violence, sexual and self-harm content